

EIE6207: Estimation Theory

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References:

- Steven M. Kay, *Fundamentals of Statistical Signal Processing*, Prentice Hall, 1993.
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Overview

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- Estimation in signal processing is fundamental to many applications:
 - Radar
 - Sonar
 - Speech and audio processing
 - Image and video processing
 - Biomedical and biomedicine
 - Communications
 - Control engineering
 - Seismology
- All of these applications require to *estimate* a set of *parameters*, e.g., the position of aircraft through radar, position of a submarine through sonar, and phonemes of speech waveform.
- All of these applications have a common characteristics: The parameters are estimated from noisy signals. So, their value can only be an estimate.

A Naive Example

- Assume that you have a coin but do not know its *true* probability p of getting a head if you flip it.
- You do an experiment to flip it N times and get N_h heads.
- In statistics, we say that you have an estimator $\hat{p} = \frac{N_h}{N} \approx p$.
- The approximation becomes very close when N is big.
- An estimator is a way of estimating an unknown parameter (p in our example) of a statistical model (the coin) given some data (no. of heads in N trials).

Mean Estimator

- In all of these applications, we estimate the values of parameters based on some waveforms.
- Specifically, Given N discrete-time $\{x[0], x[1], \dots, x[N-1]\}$ that depends on an unknown parameter θ , we define an *estimator* $\hat{\theta}$ to estimate θ :

$$\hat{\theta} = g(x[0], x[1], \dots, x[N-1]),$$

where g is some function.

- For example, for the mean estimator, $g(\cdot)$ is given by

$$g(x[0], x[1], \dots, x[N-1]) = \frac{1}{N} \sum_{i=0}^{N-1} x[i]$$

Estimator of a Linear Model

- The stock price in the figure below seems to increase on average.
- To verify this hypothesis, we use the model:

$$x[n] = A + Bn + w[n], \quad n = 0, 1, \dots, N - 1,$$

where $w[n]$ is white Gaussian noise with PDF $\mathcal{N}(0, \sigma^2)$.



Figure: y-axis: stock price $x[n]$; x-axis: time point n

Estimator of a Linear Model

- If $w[n]$ is uncorrelated with $x[n]$, then

$$\begin{aligned} p(\mathbf{x}; \boldsymbol{\theta}) &= \prod_{n=1}^{N-1} p(x[n]; \boldsymbol{\theta}) \\ &= \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (x[n] - A - Bn)^2 \right\} \\ &= \frac{1}{(2\pi\sigma^2)^{N/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A - Bn)^2 \right\} \end{aligned}$$

where $\mathbf{x} = [x[0] \ \dots \ x[N-1]]^T$ and $\boldsymbol{\theta} = [A \ B]^T$.

- Maximizing $p(\mathbf{x}; \boldsymbol{\theta})$ with respect to $\boldsymbol{\theta}$, we obtain close-form solutions for A and B .
- For the diagram shown in previous page, $B > 0$, meaning the stock price is increasing.

Bayesian Estimator

- So far, we assume that θ is deterministic but unknown. This is called *classical* estimation.
- But we could also bring in prior knowledge about θ , e.g., prior range or distribution of $\theta = [A \ B]^T$. Then, we have a *Bayesian* estimator.
- The data and parameters are modeled by a joint PDF:

$$p(\mathbf{x}, \theta) = p(\mathbf{x}|\theta)p(\theta),$$

where $p(\theta)$ is the prior PDF, which summarizes our knowledge about θ before any data are observed.

- $p(\mathbf{x}|\theta)$ is the conditional PDF, summarizing our knowledge about $\mathbf{x}$ when we know θ .

Minimum Variance Unbiased Estimation

Minimum Variance Unbiased Estimation

- An estimator is unbiased if *on average* it yields the true value of the unknown parameter θ .
- Mathematically, an estimator is unbiased if

$$\mathbb{E}(\hat{\theta}) = \theta \quad a < \theta < b.$$

- Example: Unbiased estimator for DC level with white Gaussian noise (WGN):
 - Consider the observations

$$x[n] = A + w[n] \quad n = 0, 1, \dots, N-1 \quad (1)$$

where $-\infty < A < \infty$ is the parameter to be estimated and $w[n]$ is WGN.

- A reasonable estimator for the average value of $x[n]$ is

$$\hat{A} = \frac{1}{N} \sum_{n=0}^{N-1} x[n]$$

Minimum Variance Unbiased Estimation

- Using the linearity property of expectation:

$$\mathbb{E}(\hat{A}) = \frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E}(x[n]) = A.$$

- The sample mean is an unbiased estimator of the DC level in WGN.
- If multiple estimators of the same parameter θ are available, i.e., $\{\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_K\}$, a better estimator can be obtained by averaging these estimators:

$$\hat{\theta} = \frac{1}{K} \sum_{k=1}^K \hat{\theta}_k.$$

- If all estimators are unbiased, we have

$$\mathbb{E}(\hat{\theta}) = \theta$$

- If all estimators have the same variance, then

$$\text{var}(\hat{\theta}) = \frac{1}{K^2} \sum_{k=1}^K \text{var}(\hat{\theta}_k) = \frac{\text{var}(\hat{\theta}_1)}{K} \implies \text{variance decrease}$$

Minimum Variance Unbiased Estimation

- However, a *biased* estimator is characterized by a systematic error, e.g., $\mathbb{E}(\hat{\theta}_k) = \theta + b(\theta)$. Then,

$$\mathbb{E}(\hat{\theta}) = \frac{1}{K} \sum_{k=1}^K \mathbb{E}(\hat{\theta}_k) = \theta + b(\theta)$$

- This means that having more *biased* estimators does not mean that the bias can be reduced.

Minimum Variance Criterion

- Mean square error (MSE):

$$\text{mse}(\hat{\theta}) = \mathbb{E} \left\{ (\hat{\theta} - \theta)^2 \right\}$$

- The MSE has problem if the estimator is biased (see Tutorial Q1):

$$\begin{aligned} \text{mse}(\hat{\theta}) &= \mathbb{E} \left\{ \left[\left(\hat{\theta} - \mathbb{E}(\hat{\theta}) \right) + \left(\mathbb{E}(\hat{\theta}) - \theta \right) \right]^2 \right\} \\ &= \text{var}(\hat{\theta}) + \left[\mathbb{E}(\hat{\theta}) - \theta \right]^2 \\ &= \text{var}(\hat{\theta}) + b^2(\theta) \end{aligned} \tag{2}$$

- This means that the MSE comprises errors due to the variance of the estimators and the bias.
- Therefore, we should find an unbiased estimator (with $b = 0$) that produces the minimum variance.

Example of Biased Estimators

- Let the estimator of A in Eq. 1 be

$$\tilde{A} = \frac{\alpha}{N} \sum_{n=0}^{N-1} x[n],$$

where α is an arbitrary constant.

- We would like to find the value of α that minimize

$$\begin{aligned} \text{mse}(\tilde{A}) &= \mathbb{E} \left\{ (\tilde{A} - A)^2 \right\} = \text{var}(\tilde{A}) + \left[\mathbb{E}\{\tilde{A}\} - A \right]^2 \\ &= \frac{\alpha^2 \sigma^2}{N} + (\alpha - 1)^2 A^2 \quad (\text{See Tut Q2}) \end{aligned}$$

- Setting $\frac{\partial \text{mse}(\tilde{A})}{\partial \alpha} = 0$, we obtain

$$\alpha^* = \frac{A^2}{A^2 + \sigma^2/N}$$

- The optimal value α^* depends on the unknown parameter A .
Therefore the estimator is not realizable.

Example of Biased Estimators

- Given a dataset, its sample variance is a biased estimator of the true population variance (see Tut Q3).

- We could focus on estimators that have zero bias so that the bias contributes nothing to the MSE.
- Without the bias term, the MSE in Eq. 2 does not involve the unknown parameter.
- By focusing on estimators with zero bias, we may hope to arrive at a design criterion that yields realizable estimators.
- **Definition:** An estimator $\hat{\theta}$ is called unbiased if its bias is zero for all values of the unknown parameter.
- **Recap:** For an estimator to be unbiased, we require that on average the estimator will yield the true value of the unknown parameter.

- Some unbiased estimators are more useful than others. For example, consider two estimators for Eq. 1:

$$\hat{A}_1 = x[0] \text{ and } \hat{A}_2 = \frac{1}{N} \sum_{n=0}^{N-1} x[n].$$

Although $\mathbb{E}(\hat{A}_1) = \mathbb{E}(\hat{A}_2) = A$, $\hat{A}_2$ is better because

$$\text{var}(\hat{A}_1) = \sigma^2 > \text{var}(\hat{A}_2) = \frac{\sigma^2}{N}. \quad (3)$$

How Good is Our Estimator

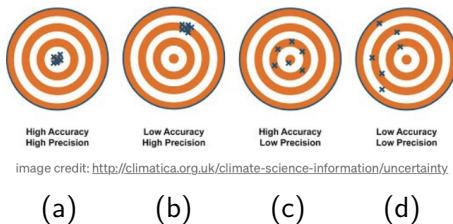


image credit: <http://climatica.org.uk/climate-science-information/uncertainty>

- If you hit the bullseye on average, you are an unbiased dart-thrower, e.g., (a) and (c).
- If your throws often off-centered, you are a biased dart-thrower, e.g., (b) and (d).
- A *good* unbiased dart-thrower should hit the bullseye on average with *small* variation around the bullseye, e.g., (a).
- A *bad* unbiased dart-thrower has large variation, although he/she can hit the bullseye on average, e.g., (c).
- A really *bad* biased dart-thrower has large variation around a point with an offset from the bullseye, e.g., (d).

Minimum Variance Unbiased Estimators

- If the bias is zero, the MSE is just the variance.
- This gives rise to the minimum variance unbiased estimator (MVUE) for θ .
- **Definition:** An estimator $\hat{\theta}$ is the MVUE if it is (1) *unbiased* and (2) has the *smallest variance* among any unbiased estimator for all values of the unknown parameter θ , i.e., $\forall \hat{\theta}$, we have

$$\text{var}(\hat{\theta}_{\text{MVUE}}) \leq \text{var}(\hat{\theta})$$

- It is important to note that a uniformly MVUE may not always exist, and even if it does, we may not be able to find it.
- This definition and the above discussions apply to vector parameters θ .

Cramer-Rao Lower Bound

Cramer-Rao Lower Bound

- Eq. 2 suggests that the variance of an unbiased estimator is the MSE, i.e., if $b(\theta) = 0$, we have

$$\text{mse}(\hat{\theta}) = \text{var}(\hat{\theta})$$

- But how do we know that the unbiased estimator $\hat{\theta}$ is the one with the smallest MSE?
- Also, how do we compute its variance?
- This quantity is given by the *Cramer-Rao lower bound (CRLB)*.
- The CRLB theorem induces a lower bound on the MSE of an estimator
- The CRLB theorem also provides a method for finding the best estimator.

Cramer-Rao Lower Bound

- We will use the example in Eq. 1 to explain the idea of CRLB.
- For simplicity, suppose that we are using only a single observation $x[0]$ to estimate A in Eq. 1, $x[n] = A + w[n]$.
- The PDF of $x[0]$ is

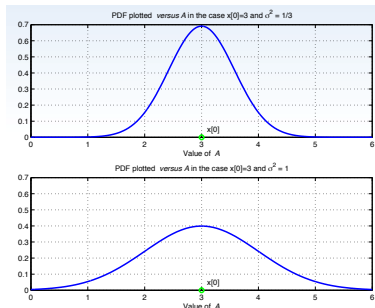
$$p(x[0]; A) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (x[0] - A)^2 \right\}$$

- Once we have observed $x[0]$, say for example $x[0] = 3$, some values of A are more likely than others.

Cramer-Rao Lower Bound

- The pdf of A has the same form as the PDF of $x[0]$:

$$\text{pdf of } A = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (3 - A)^2 \right\}$$



- We probably will get more accurate estimate of A for the distribution of A in the upper panel.

Cramer-Rao Lower Bound

- If the pdf is viewed as a function of the unknown parameter (with x fixed), it is called the *likelihood function* (function of the unknown parameters).
- The “sharpness” of the likelihood function determines how accurately we can estimate the parameter.
- If a single sample is observed as $x[0] = A + w[0]$, then we can expect a better estimate if σ^2 is small.
- How to quantify the sharpness? Is there any measure that would be common for all possible estimators for a specific estimation problem?
- The second derivative of the likelihood function (or log-likelihood) is an alternative for measuring the sharpness of a function.

Cramer-Rao Lower Bound

- Recall that the PDF of $x[0]$ is

$$p(x[0]; A) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (x[0] - A)^2 \right\}$$

- The log-likelihood function:

$$\log p(x[0]; A) = -\log \sqrt{2\pi\sigma^2} - \frac{1}{2\sigma^2} (x[0] - A)^2$$

- The first derivative w.r.t. A is the score function:

$$\nabla_A \log p(x[0]; A) = \frac{\partial \log p(x[0]; A)}{\partial A} = \frac{1}{\sigma^2} (x[0] - A)$$

- The second derivative is independent of $x[0]$:

$$\frac{\partial^2 \log p(x[0]; A)}{\partial A^2} = -\frac{1}{\sigma^2}$$

Cramer-Rao Lower Bound

- Because σ^2 is the smallest possible variance, for this particular case, we have an alternative way of finding the minimum variance:

$$\text{minimum variance} = \sigma^2 = \frac{1}{-\frac{\partial^2 \log p(x[0]; A)}{\partial A^2}}$$

- For the general cases,
 - If the function depends on data $\mathbf{x}$, take the expectation over all $\mathbf{x}$'s.
 - If the function depends on the parameter θ , evaluate the derivative at the true value of θ ,
- We have the general rule:

$$\text{min variance of any unbiased estimator} = \frac{1}{-\mathbb{E}_{\mathbf{x} \sim p(\mathbf{x}; \theta)} \left[\frac{\partial^2 \log p(\mathbf{x}; \theta)}{\partial \theta^2} \right]},$$

CRLB Theorem

- If the pdf $p(\mathbf{x}; \theta)$ satisfies the “regularity” condition (see Tut Q5):

$$\mathbb{E}_{\mathbf{x} \sim p(\mathbf{x}; \theta)} \left\{ \frac{\partial \log p(\mathbf{x}; \theta)}{\partial \theta} \right\} = 0, \quad \forall \theta,$$

then, the variance of any unbiased estimator $\hat{\theta}$ must satisfy

$$\text{var}(\hat{\theta}) \geq \frac{1}{-\mathbb{E}_{\mathbf{x} \sim p(\mathbf{x}; \theta)} \left[\frac{\partial^2 \log p(\mathbf{x}, \theta)}{\partial \theta^2} \right]},$$

where the derivative is evaluated at the true value of θ .

- An unbiased estimator attains the bound for all θ if and only if

$$\frac{\partial \log p(\mathbf{x}; \theta)}{\partial \theta} = I(\theta)(g(\mathbf{x}) - \theta) \quad (4)$$

for some functions g and I . The MVU estimator is $\hat{\theta} = g(\mathbf{x})$ and the minimum variance $= 1/I(\theta)$.

CRLB Example 1: Estimation of DC Level in WGN

- Example: DC level in Gaussian noise:

$$x[n] = A + w[n], \quad n = 0, 1, \dots, N - 1$$

What is the minimum variance of any unbiased estimator using N samples?

- The likelihood function is the product of N densities:

$$\begin{aligned} p(\mathbf{x}; A) &= \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (x[n] - A)^2 \right\} \\ &= \frac{1}{(2\pi\sigma^2)^{N/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2 \right\} \end{aligned}$$

CRLB Example 1: Estimation of DC Level in WGN

- The log-likelihood function is

$$\log p(\mathbf{x}; A) = -\log \left[(2\pi\sigma^2)^{\frac{N}{2}} \right] - \frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2$$

- The first derivative is

$$\frac{\partial \log p(\mathbf{x}, A)}{\partial A} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x[n] - A) = \frac{N}{\sigma^2} (\bar{x} - A),$$

where $\bar{x}$ is the sample mean.

- The 2nd derivative is

$$\frac{\partial^2 \log p(\mathbf{x}, A)}{\partial A^2} = -\frac{N}{\sigma^2}$$

- Therefore, the minimum variance of any unbiased estimator is

$$\text{var}(\hat{A}) \geq \frac{\sigma^2}{N} \quad (5)$$

CRLB Example: Estimation of DC Level in WGN

- In Eq. 3, the sample mean A_2 gives the minimum variance in Eq. 5. Therefore, the sample mean is an MVUE for this problem.
- Using the CRLB theorem in Eq. 4, we have

$$\begin{aligned}\frac{\partial \log p(\mathbf{x}; A)}{\partial A} &= \frac{N}{\sigma^2}(\bar{x} - A) \\ &= I(A)(g(\mathbf{x}) - A),\end{aligned}$$

where $I(A) = \frac{N}{\sigma^2}$ and $g(\mathbf{x}) = \bar{x}$.

- So, $g(\mathbf{x})$ is the MVUE and $1/I(A)$ is its variance.

- Cramer Rao inequality provides lower bound for the estimation error variance.
- Minimum attainable variance is often larger than CRLB.
- We need to know the pdf to evaluate CRLB. If the data is multivariate Gaussian or i.i.d. with known distribution, we can evaluate it.
- Often we don't know this information and cannot evaluate this bound.
- It is not guaranteed that MVUE exists or is realizable.